

Generalized Contraction Framework

GENERALIZED CONTRACTION THEOREM (Hussain-Sehgal)

Let (X, d) be a complete metric space, $T: X \rightarrow X$

Define $\phi: (\mathbb{R}^+)^5 \rightarrow \mathbb{R}^+$ continuous, nondecreasing in each variable

Suppose T satisfies:

$$d(Tx, Ty) \leq \phi(d(x, Tx), d(y, Ty), d(x, Ty), d(y, Tx), d(x, y))$$

If $\phi(t, t, a_1t, a_2t, a_5t) < t$ for all $t > 0$

where $a_1, a_2, a_5 \in \{0, 1, 2\}$ with $a_1 + a_2 \leq 2$

Then: T has unique fixed point u in X

$$\lim(T^n x) = u \text{ for any } x \in X$$

SPECIAL CASES:

- Banach: $\phi(\cdot, \cdot, \cdot, \cdot, d(x, y)) = k \cdot d(x, y)$, $k < 1$
- Kannan: $\phi(a, b, \cdot, \cdot, 0) = r(a+b)/2$, $r < 1/2$
- Ćirić (HR): $\phi = a_1t_1 + a_2t_2 + a_3t_3 + a_4t_4 + a_5t_5$
with $a_1+a_2+a_3+a_4+a_5 < 1$